

Academic Research Profile

Master's and PhD Research Trajectory

This document summarizes the progression from the Master's research in deep-learning-based medical image classification to the current PhD research direction. The central continuity is the design and structuring of deep-learning models to improve class-wise performance consistency while maintaining trustworthy predictive behavior.

1. Master's Research

Master's research focused on deep-learning-based image classification for Wireless Capsule Endoscopy (WCE), with particular emphasis on improving classification performance through model optimization and systematic experimentation.

Main focus areas:

- EfficientNet-based deep-learning architectures
- Data augmentation strategies
- Model optimization and hyperparameter tuning
- Ensemble learning
- Medical image classification and evaluation

A key outcome of this work was the recognition that aggregate accuracy alone does not fully describe model behavior: a model can achieve strong overall performance while showing substantial differences in performance between individual classes.

2. Transition to PhD Research

The PhD research develops this observation into a broader model-design problem. Instead of focusing only on maximizing overall classification accuracy, the research asks how the internal structure and mechanisms of deep-learning models can be designed to produce more consistent class-wise performance together with reliable and trustworthy predictions.

3. Central PhD Research Question

How can deep-learning models, particularly CNN-based architectures, be structured and designed to achieve more consistent performance across classes while maintaining trustworthy and reliable predictions, especially under limited-data conditions?

4. Main Research Dimensions

4.1 Class-wise Performance Consistency

The first core dimension concerns the consistency of predictive performance across classes. The objective is to understand and reduce disparities in class-wise accuracy and related performance measures, rather than relying exclusively on aggregate metrics.

4.2 Trustworthy Predictive Performance

The second dimension concerns whether model predictions can be relied upon. This includes the study of prediction confidence, uncertainty, calibration, robustness, and the ability to identify situations in which the model may be unreliable.

4.3 Model and Representation Structure

The research therefore examines how architectural components and learned representations can be structured to support both consistency and trustworthy behavior. Attention mechanisms and feature representations are potential intervention points.

5. Information-Theoretic Principles as a Research Direction

Information-theoretic methods are not the starting point of the PhD problem; they are a potential methodological direction for addressing it. Entropy and related information measures can be investigated as mechanisms for characterizing or controlling feature representations, attention, uncertainty, and information distribution within deep-learning models.

The current entropy-constrained channel-attention work represents an initial investigation of this direction. The broader PhD question remains independent of any single method, allowing other architectural, uncertainty-aware, or representation-learning approaches to be evaluated as the research develops.

6. Overall Research Trajectory

Stage	Primary Focus	Research Question / Outcome
Master's	WCE image classification	How can deep-learning models be optimized to improve medical image classification performance?
PhD — Core Problem	Class-wise consistency + trustworthiness	How can model structure be designed to provide more consistent class-wise performance and reliable predictions?
PhD — Methodological Direction	Information-theoretic principles	Can entropy and related information measures provide principled mechanisms for improving representation, consistency, and trustworthy behavior?
Potential Extensions	Broader visual perception	Can the developed principles extend beyond classification to detection, segmentation, and other visual-perception tasks?

7. Short Research Identity

The research trajectory can be summarized as the development of deep-learning models that are not only accurate, but also more consistent across classes and more trustworthy in their predictions. The work begins from practical deep-learning and computer-vision problems and progressively investigates model structure, representation learning, uncertainty, and information-theoretic principles as possible mechanisms for achieving reliable visual perception under limited-data conditions.